

9. A spacecraft is travelling at $6.0 \times 10^7 \text{ m s}^{-1}$ relative to a star.

An observer on the spacecraft measures the speed of light emitted by the star to be

A $2.4 \times 10^8 \text{ m s}^{-1}$

B $2.9 \times 10^8 \text{ m s}^{-1}$

C $3.0 \times 10^8 \text{ m s}^{-1}$

D $3.1 \times 10^8 \text{ m s}^{-1}$

E $3.6 \times 10^8 \text{ m s}^{-1}$.